

Real-Time Questions for All Distributions

BINOMIAL DISTRIBUTION:

1. 95% defect-free items; probability exactly 9 good out of 10.
2. Student solves 60% problems; probability solves at least 3 of 5.
3. Basketball free-throw success 70%; probability makes 6 of 8.
4. Quiz guessing 25%; probability 4 correct out of 10.
5. Call center 80% resolution; probability 7 satisfied from 10.

POISSON DISTRIBUTION:

1. Website gets 6 visits/min; probability exactly 10 next minute.
2. Bank gets 3 customers/5 min; probability 5 in next 5 min.
3. Hospital gets 10 patients/hr; probability >12 next hour.
4. Call center gets 12 calls/hr; probability no calls in 5 min.
5. Toll booth sees 4 cars/min; probability 2 cars in 30 sec.

CONFIDENCE INTERVAL:

1. 50 students, mean=165, SD=10; 95% CI.
2. Bottle fill mean=500ml, SD=8, n=40; 99% CI.
3. Salaries mean=25000, SD=4000, n=30; 95% CI.
4. Restaurant rating mean=4.2, SD=0.5, n=80; 90% CI.
5. Exam mean=70, SD=12, n=100; 95% CI.

NORMAL DISTRIBUTION:

1. Heights mean=170, SD=10; probability >180.
2. IQ mean=100, SD=15; probability 90–110.
3. Weight mean=65, SD=8; probability <60.
4. Exam mean=70, SD=12; probability >85.
5. Bulb life mean=1000, SD=80; probability <900.

EXPONENTIAL DISTRIBUTION:

1. Customer arrives every 10 min; probability within 3 min.
2. System crash avg 5 hrs; probability crash within 1 hr.
3. ATM wait avg 4 min; probability >6 min.
4. Email every 2 min; probability next after 3 min.
5. Elevator avg 8 min; probability within 5 min.

UNIFORM DISTRIBUTION:

1. Lucky draw 1–100; probability 30–60.
2. Bus arrival 0–20 min; probability <5 min.
3. Game score 10–50; probability >40.

4. Arrival time 8–9 AM; probability before 8:20.
5. Sensor reading 5–25; probability 10–15.